# Using the BL5A GUI

This is the operator's guide to the window: what each part shows, what each button sends, and the two experiments from start to end. To build the programs and set them up at the beamline, see install.md.

The screenshots were taken against the simulation (`bl5a-sim`), so the ring current, shutter and temperature fields read "—" in them.

## Before you start

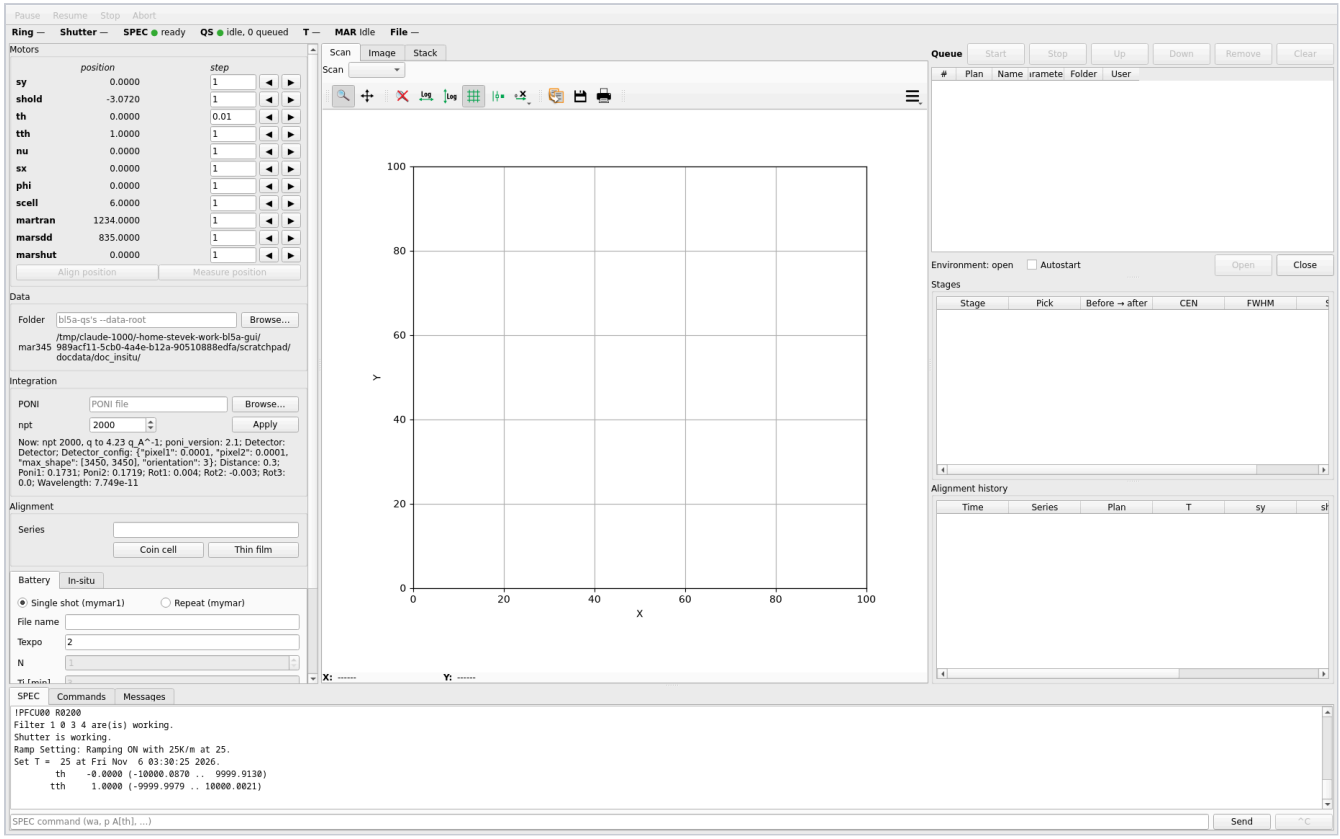
The GUI is a client. On its host, user systemd keeps spec-ioc, `bl5a-qs` (the queue server, which runs the plans) and `bl5a-integrator` (which integrates each frame) running (install.md, section 5); the beamline runs the mar345 IOC. You start SPEC in server mode, then the GUI with the **BL5A** shortcut on the desktop or in the applications menu. `systemctl --user status 'bl5a*'` shows whether the services run.

In a terminal, `deploy/bl5a-gui` starts the same window and takes these options:

| Option | Default | What it is |
|---|---|---|
| `--qs-control` | `tcp://localhost:60715` | bl5a-qs control socket |
| `--qs-documents` | `tcp://localhost:60725` | bl5a-qs document stream |
| `--integrator-rep` | `tcp://localhost:60735` | integrator requests |
| `--integrator-pub` | `tcp://localhost:60745` | integrator new rows and errors |
| `--user` | `bl5a` | the name in the queue's User column |

You can close the GUI at any time. The queue keeps running in bl5a-qs and the stacks stay with the integrator. A new window starts with empty Scan, Stages, Alignment history, Commands and Messages panes: they are filled from what the plans report while the window is open. The Stack tab asks the integrator, so it shows every stack it holds.

# The window




*The window at rest*

| Area | What is in it |
|---|---|
| Top | the run buttons (Pause, Resume, Stop, Abort) and the status line |
| Left | motors, the Data and Integration boxes, the alignment buttons, the Battery and In-situ forms |
| Centre | the Scan, Image and Stack tabs |
| Right | the queue, the stages of the running alignment, the alignment history |
| Bottom | the SPEC console, the Commands the plans sent, the GUI's Messages |

The panes are split by handles you can drag.

## Who drives SPEC

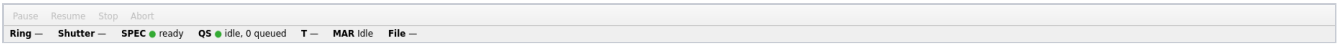
SPEC runs one command at a time. While the queue runs a plan, the plan owns SPEC, so the GUI greys out everything that would send SPEC a command of its own: the motor tweak buttons, their step fields and the SPEC console's entry.

They are live only when all of these hold:

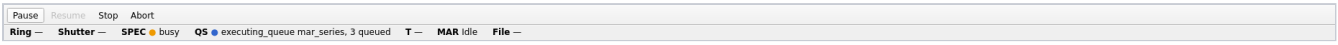
- bl5a-qs is idle and runs nothing;
- spec-ioc reports SPEC connected (`BL5A:SPEC:Connected`);
- SPEC is ready (`BL5A:SPEC:Busy`);
- no console command is running (`BL5A:SPEC:Cmd:Busy`).

The queue itself can be edited at any time.

## Status line




*The status line at rest*




*The status line while a mar_series runs*

| Field | Shows |
|---|---|
| Ring | the ring current, mA (`BL5A:userCalcOut1.VAL`) |
| Shutter | the front-end shutter; its PV is not configured yet, so "—" |
| SPEC | ● ready (green), ● busy (orange), ● disconnected (red); "—" when spec-ioc does not answer |
| QS | the queue server's state, the running plan, how many items wait; "environment closed" and "stopping after the frame" when they apply; ● not answering (red) when bl5a-qs does not reply |
| T | the temperature as readback → setpoint, and the ramp; "—" until the temperature PVs are configured |
| MAR | the mar345 detector state (`DetectorState_RBV`) |
| File | the full path of the last frame; you can select and copy it |

The QS dot is green when idle, orange when paused and blue otherwise.

## Run buttons

| Button | Live when | What it does |
|---|---|---|
| Pause | a plan runs | pauses at once; a moving motor is stopped |
| Resume | paused | goes on |
| Stop | a plan runs or is paused | ends the plan, recorded as a success |
| Abort | a plan runs or is paused | asks first, then ends the plan, recorded as aborted |

However a plan ends, bl5a-qs then interrupts a SPEC command still running, stops a count still running and closes the mar345 shutter.

The plans do not replay a SPEC macro that was cut in half. After **Pause**, **Resume** fails the plan with "plan interrupted: a paused plan cannot resume, abort it"; end it with **Stop** or **Abort**. To end a battery series cleanly, use **Stop after frame** on the Battery form instead: that pause lands between frames, where the series can go on.

A motor's stop reaches SPEC as `abort_all`, which stops every motor SPEC is moving.

The queue's own **Stop** is different: it lets the running item finish and then stops the queue.

## Motors



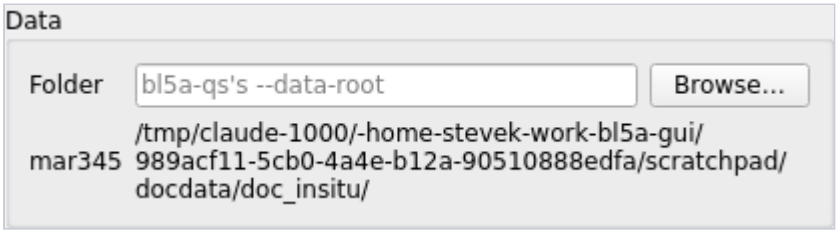

*The motors panel*

One row per spec-ioc motor: `sy shold th tth nu sx phi scell martran marsdd marshut`.

- **position** is the readback, with the record's precision and unit.
- An orange dot after it means the motor is moving.
- **step** is the tweak step (the motor record's `TWV`). Type a positive number and leave the field (Tab or click elsewhere) to set it.
- ◀ moves the motor by −step, ▶ by +step. spec-ioc turns each into a SPEC move.

**Align position** and **Measure position** stay disabled until BL5A gives these positions.

## Data




*The Data box*

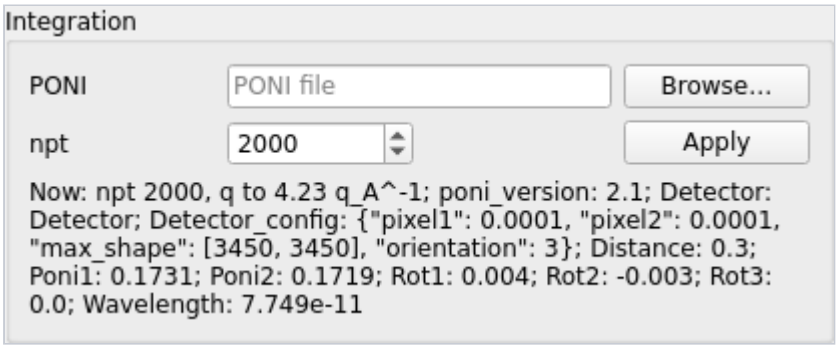
Everything an item records goes in one folder, `<data root>/<series>/`: its frames, its scans as the SPEC file `<series>.spec` and its integrated stacks. bl5a-qs makes the folder when the item starts. The series is the one typed in the form, or the plan's name when there is none, so the items of one series share a folder. Type the same series in **Alignment** and on the **Battery** form and a cell's alignment scans and frames land together:

```
<data root>/
  cell7/
    cell7.spec              the align_coin scans
    cell7_0001.mar3450 …    the frames
    cell7.h5                the stack
  film_A/
    film_A.spec             each in-situ row's alignment scans
    filmA_30_0001.mar3450 …
    film_A.h5
```

- **Folder** is the data root of the items queued from now on; empty means bl5a-qs's `--data-root`. **Browse…** picks it, and the GUI remembers it. It must be an absolute path without spaces that is the same on the bl5a-qs, mar345 IOC and integrator hosts (see install.md).
- Each queued item carries the folder as it was when queued, so a change does not move items already in the queue. The queue's **Folder** column shows where each item goes.
- **mar345** is the folder the mar345 IOC writes to now (`FilePath_RBV`); bl5a-qs sets it before each frame. A red "missing" after it means the IOC does not see that folder (`FilePathExists_RBV` reads No).

A series is one folder name without spaces, `/` or control characters, and not `.` or `..`; the whole folder, `<data root>/<series>`, is at most 200 bytes. The mar345 detector server takes a frame's path as one word, so a space would split it. The forms refuse any other series, and queueing refuses a **Folder** with a space or a path that is too long. In the stack's file name, a character other than a letter, digit, `_`, `-` or `.` is written `%XX`: series `LiFePO4(1C)` gives `LiFePO4%281C%29.h5`. An item whose data root does not exist fails when it starts, with "data folder: … the data root … does not exist".

## Integration




*The Integration box*

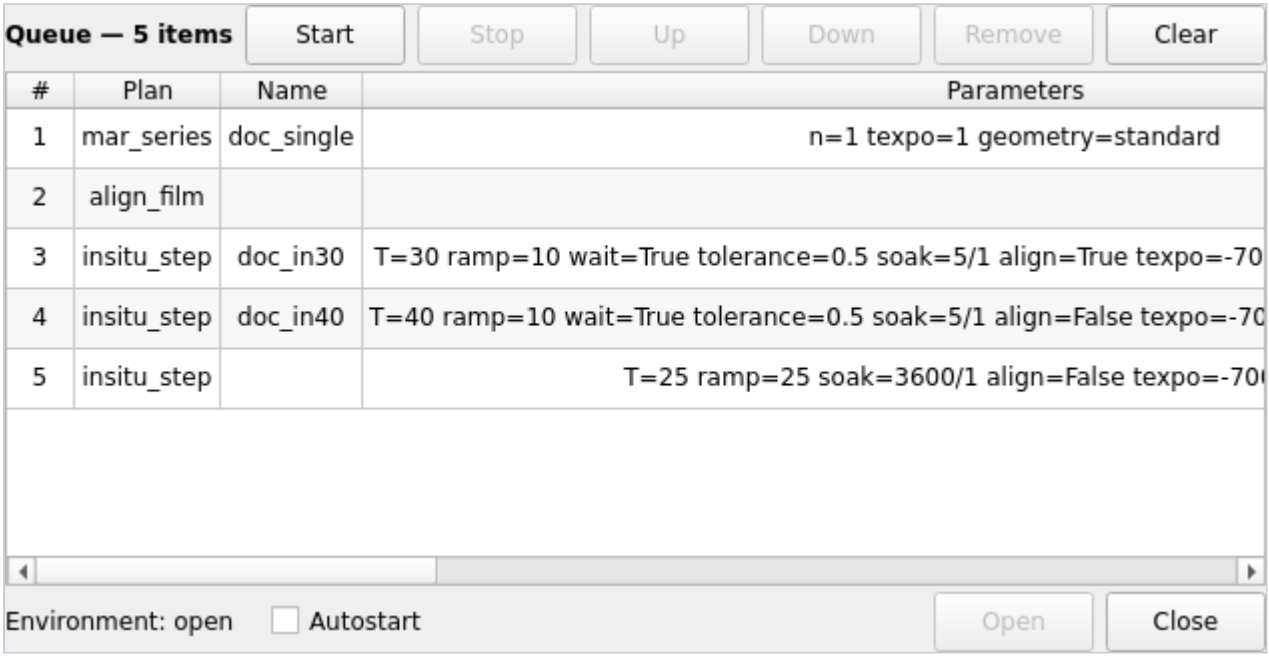
The text under the buttons is what the integrator integrates with now: npt, the q range and the PONI.

| Control | What it is |
|---|---|
| PONI | a pyFAI PONI file for the full 345 mm plate; **Browse…** picks it |
| npt | the bins from q = 0 to the plate's far rim (default 2000) |
| Apply | reads the file and sends its text and npt to the integrator |

The integrator checks them first: a PONI it cannot use is refused, and the box adds "Refused:" and the reason. An accepted one applies to the frames after those already waiting to be integrated; until then the box shows it as "Next, after the queued frames". When it takes effect, Messages says "integrator: npt …, q to … from now on" and the Image tab's rings move to the new PONI.

A new npt, or a PONI that moves the q range, starts a new stack: the next frame of `cell7` goes to `cell7~2.h5`, and `cell7.h5` keeps its q axis. The Stack tab lists both. The integrator keeps the setting until it restarts, and then starts again from its `--poni` and `--npt`.

## Queue




*The queue with five items*

The top row, marked ▶ on a green background, is the running item; the rest wait in order.

| Column | Shows |
|---|---|
| # | the place in the queue |
| Plan | `mar_series`, `align_coin`, `align_film` or `insitu_step` |
| Name | the frame file name, when the plan takes frames |
| Parameters | the item's arguments, for example `T=30 ramp=10 wait=True tolerance=0.5 soak=5/1 align=True texpo=-70000 th_list=-0.05/0.11/0.13/0.15` |
| Folder | where the item's data goes: `<data root>/<series>`, or the series alone when the item goes to bl5a-qs's `--data-root` |
| User | the `--user` that queued it |

| Button | What it does |
|---|---|
| Start | runs the queue from its first item (only with items and nothing running) |
| Stop | stops the queue after the running item; that item is not interrupted |
| Up, Down | moves the selected item one place |
| Remove | takes the selected items out (also the Delete key); not the running item |

| Button | What it does |
|---|---|
| Clear | empties the queue after asking; the running item goes on |

The row under the list belongs to the queue server:

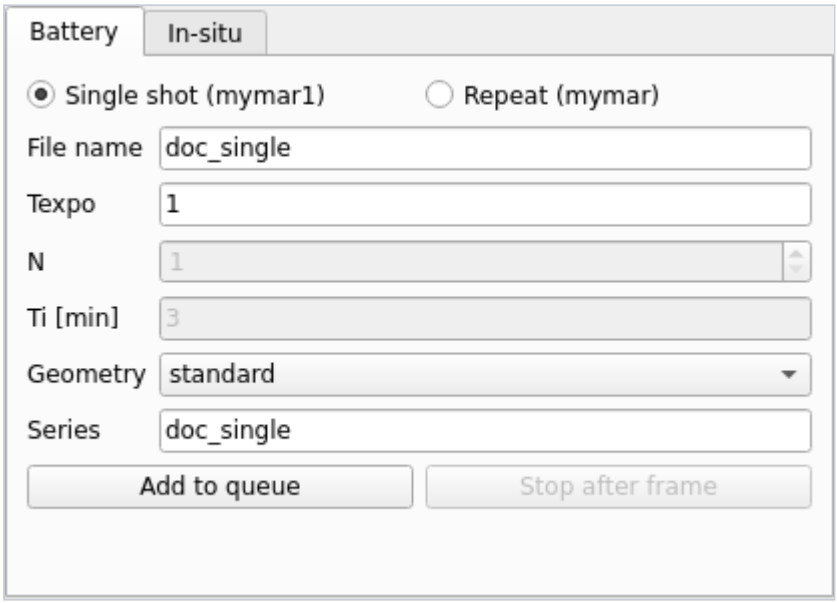
- **Environment: open / closed / not known.** Nothing runs while it is closed. **Open** opens it. **Close** is offered only while idle.
- **Autostart.** When ticked, what is added to the queue runs without a Start. The server unticks it when the queue ends in any way other than running dry: a failed or aborted plan, or a Stop. Tick it again here.

A click on a form's queue button sends one request. The rows of the In-situ table go in as one request, in table order, so nothing can slip in between them.

bl5a-qs checks an item's arguments only when the item runs. The forms check the same rules before they queue, so a bad value is caught at the form and not halfway through the night.

## Battery measurements

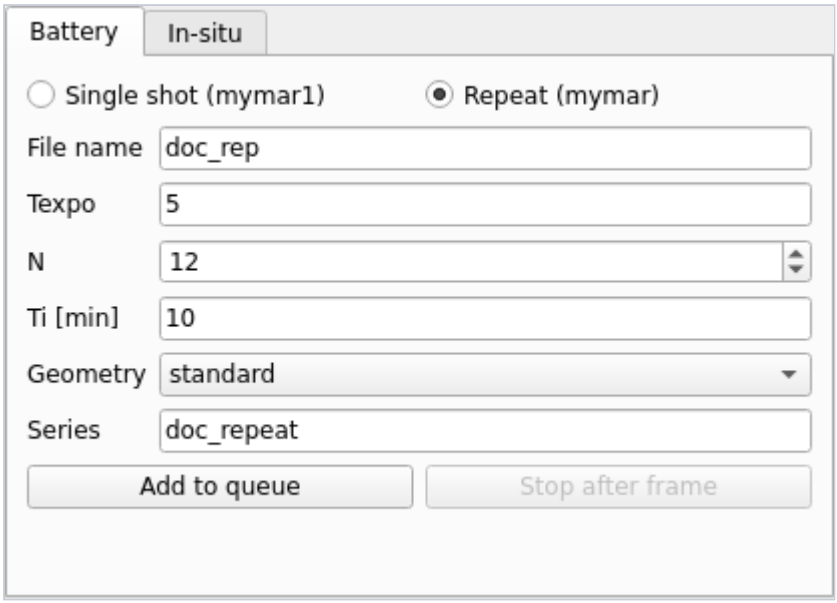
The Battery form queues `mar_series`: SPEC's `mymar1` (one frame) or `mymar` (a frame every Ti minutes).




*Single shot*

| Field | Rule | Notes |
|---|---|---|
| Single shot / Repeat | | Single shot is `mymar1`; Repeat is `mymar` |
| File name | 1 to 27 characters, no spaces or quotes | the mar345 file name |
| Texpo | seconds 1 to 20, or a negative number of monitor counts | default 2 |
| N | whole number, at least 1 | Repeat only: the number of frames |
| Ti [min] | more than 2.5 | Repeat only: from one frame's start to the next |

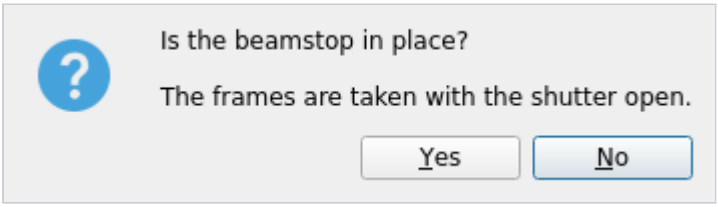
| Field | Rule | Notes |
|---|---|---|
| Geometry | `standard`, `high_q_asymm`, `low_q_symmetric` | mymar's mar345 positions, below |
| Series | a folder name (Data) | the data folder and stack the frames go to; empty means the file name |




*Repeat: twelve frames, ten minutes apart*

| Geometry | `mar_ip7` arguments |
|---|---|
| standard | `-85 435 100 330 50 1.2 -6.8` |
| high_q_asymm | `-85 435 110 330 80 1.2 -6.8` |
| low_q_symmetric | `160 435 36 345 54 1.2 -6.8` |

**Add to queue** checks the fields first. A bad value opens a "Battery" box that names the field and the rule. Then the GUI asks:




*The beamstop question*

**Yes** queues the item; **No** queues nothing.

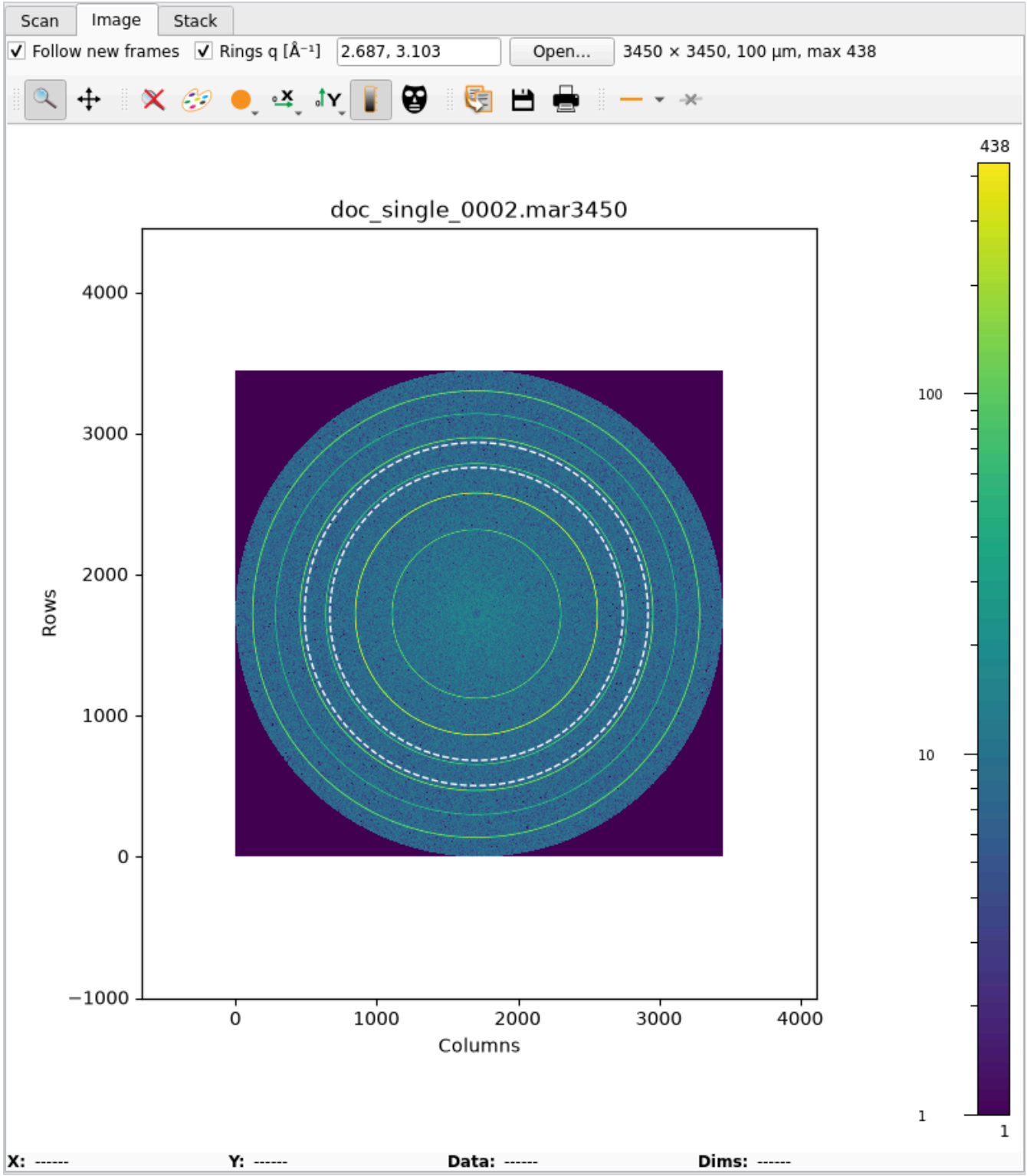
**Move tth to 80° or more first.** `mar_series` does not move tth, and `mar_ip7` refuses the upward geometry below 80° ("Two Theta Angle is too low for UPWARD Geometry."). In the simulation the plan then fails with that message; this has not been tried against the real SPEC. While the queue is idle you can type `umv tth 81` in the SPEC console. bl5a-sim does not model `umv`; there, set tth's step and click ▶ instead.

What a `mar_series` does, in order:

1. `mar345closeshutter`, `scell` to 0, `counters 2 3`, `fi 0 1 0 0`, `ic_range 4`, a 2 s wait, `mar_ip7 <geometry>`, `scell` to 0.

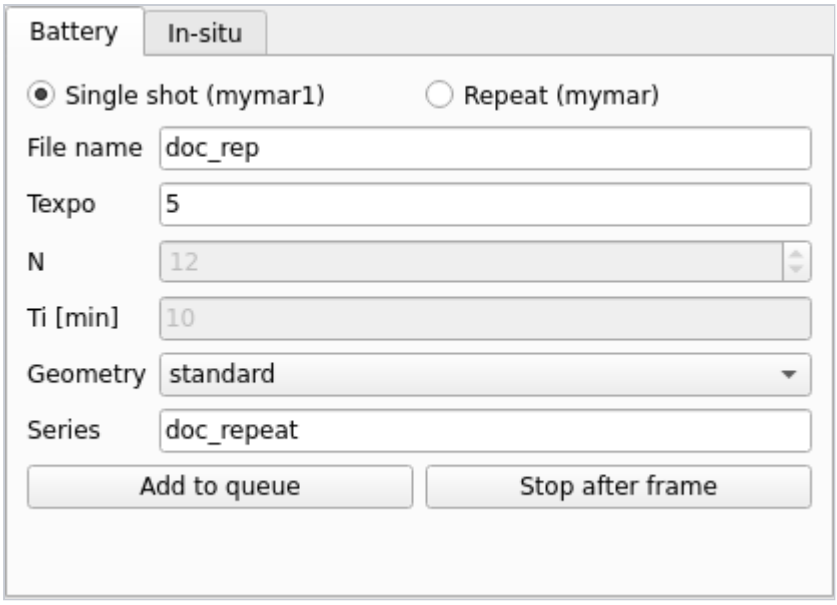
2. For each frame: the file name goes to the mar345 IOC, `mar345openshutter`, the count (in pieces of at most 429 s, or one monitor count), `mar345closeshutter`, `exp_condition`, the plate readout through the IOC, `mar345readout`.
3. Frame *i* starts *i*·Ti after the first frame. Between frames the plan only waits.

The mar345 writes `<name>_<number>.mar3450` in the item's data folder. The number is the IOC's file counter, which runs on from one name to the next.




*A single shot on the Image tab*

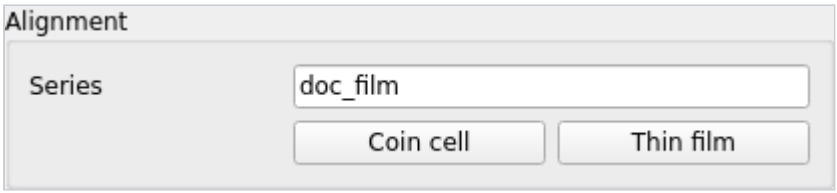
Each frame appears on the Image tab once it is read out, and is added to its series on the Stack tab once the integrator has integrated it.




*While a series runs*

**Stop after frame** is live while a `mar_series` runs. It lets the frame being taken finish, then ends the series: it asks for a pause that takes effect between frames, and sends Stop once the plan has paused. The status line says "stopping after the frame" in the meantime.

## Alignment




*The alignment buttons*

**Series** groups the runs in the alignment history, is stored with the run and names the data folder its scans go to. Empty means each run stands on its own, and its scans go to the folder `align_coin` or `align_film`.

**Coin cell** queues `align_coin`, SPEC's `coin_align`:

1. `marshut` to 0, `mon1_setup 1050 100 300 3.0 1.0`, `setkappa 50`, `counters 2 5`, `fi 1 0 0 0`, `fil 0 0 0 0`.
2. A fly scan of sy from −2.5 to 2.5 on ic1; sy goes to pl_COM.
3. A fly scan of shold from −2 to 2; shold goes to pl_COM.
4. `mar345closeshutter`, the plate is erased, `marshut` to 340.
5. The positions of sx, sy, shold, th and phi are recorded.

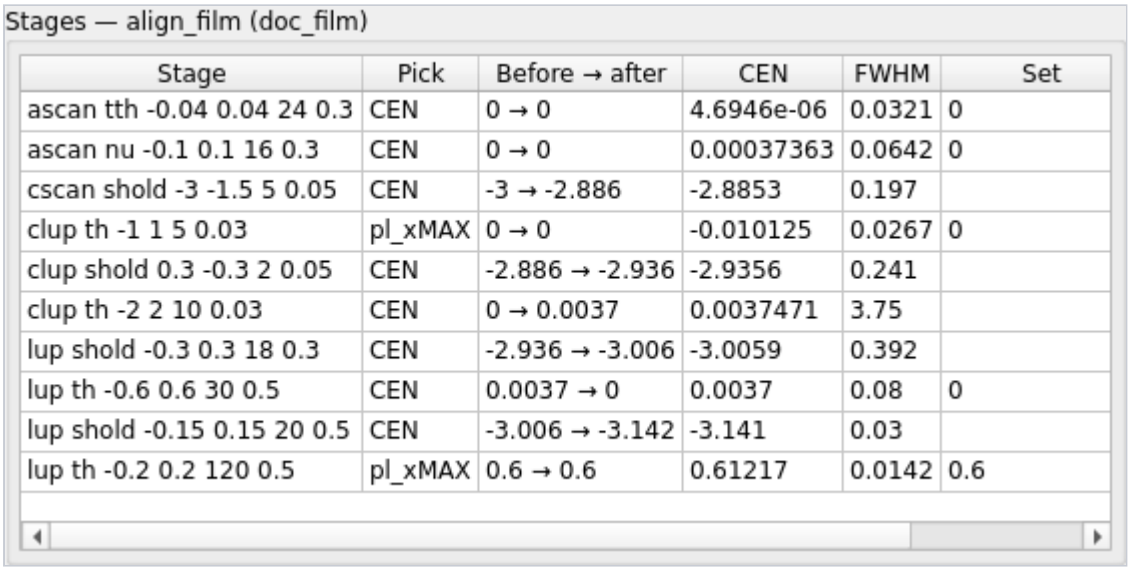
`coin_align` ends with `mar_ip` without arguments, which only prints its usage in SPEC, so the plan does not send it.

**Thin film** queues `align_film`, SPEC's `align_in.mac`. It sets the slits, attenuators and counters, moves to the direct beam (`uan 1 0`, `all6 0 0 89.7 0 0 0.0`, `shold` to −3) and then runs these stages. Each stage scans, moves the motor to the picked statistic and, where a value is given, redefines that position with `set`.

| Stage | Pick | Then |
|---|---|---|
| tth −0.04 to 0.04 | CEN | `set tth 0` |
| nu −0.1 to 0.1 | CEN | `set nu 0` |
| shold −3 to −1.5, fly | CEN | |
| th −1 to 1 relative, fly | pl_xMAX | `set th 0` |
| shold 0.3 to −0.3 relative, fly | CEN | |
| th −2 to 2 relative, fly | CEN | |
| shold −0.3 to 0.3 relative | CEN | |
| th −0.6 to 0.6 relative | CEN | `set th 0` |
| shold −0.15 to 0.15 relative | CEN | |
| `uan 1.2 0.6`, `atte 4`, then th −0.2 to 0.2 relative, 120 intervals | pl_xMAX | `set th 0.6` |

It ends with `mar345closeshutter` and an erase of the plate.

## Stages


Stages — align_film (doc_film)

| Stage | Pick | Before → after | CEN | FWHM | Set |
|---|---|---|---|---|---|
| ascan tth -0.04 0.04 24 0.3 | CEN | 0 → 0 | 4.6946e-06 | 0.0321 | 0 |
| ascan nu -0.1 0.1 16 0.3 | CEN | 0 → 0 | 0.00037363 | 0.0642 | 0 |
| cscan shold -3 -1.5 5 0.05 | CEN | -3 → -2.886 | -2.8853 | 0.197 | |
| clup th -1 1 5 0.03 | pl_xMAX | 0 → 0 | -0.010125 | 0.0267 | 0 |
| clup shold 0.3 -0.3 2 0.05 | CEN | -2.886 → -2.936 | -2.9356 | 0.241 | |
| clup th -2 2 10 0.03 | CEN | 0 → 0.0037 | 0.0037471 | 3.75 | |
| lup shold -0.3 0.3 18 0.3 | CEN | -2.936 → -3.006 | -3.0059 | 0.392 | |
| lup th -0.6 0.6 30 0.5 | CEN | 0.0037 → 0 | 0.0037 | 0.08 | 0 |
| lup shold -0.15 0.15 20 0.5 | CEN | -3.006 → -3.142 | -3.141 | 0.03 | |
| lup th -0.2 0.2 120 0.5 | pl_xMAX | 0.6 → 0.6 | 0.61217 | 0.0142 | 0.6 |

*The stages of an align_film run*

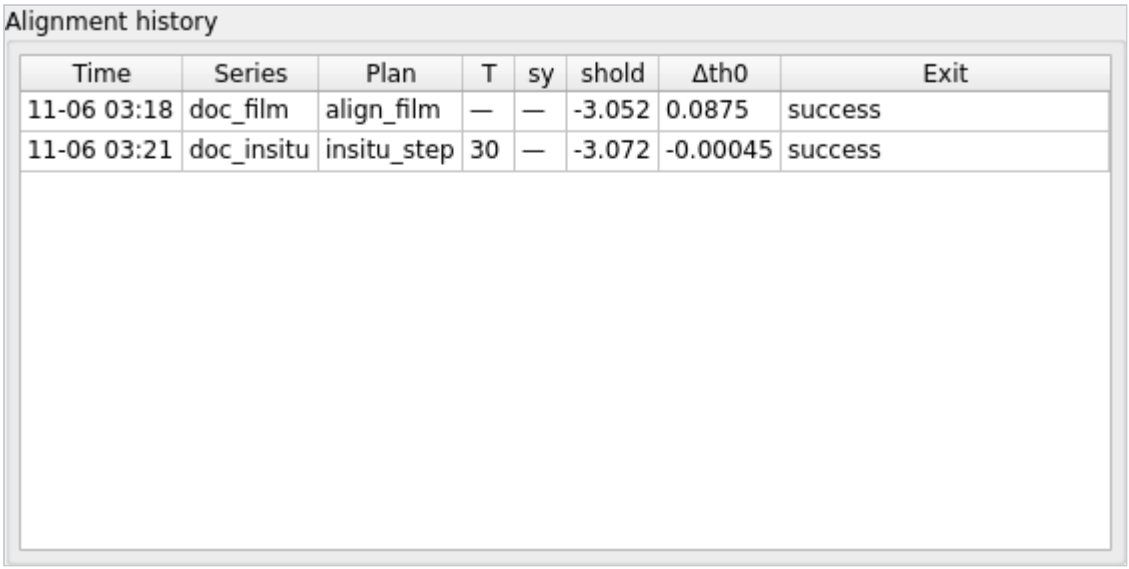
The Stages table shows each stage of the alignment running now. Its title names the plan and series. It is cleared when the next `align_coin`, `align_film` or `insitu_step` starts, including an `insitu_step` row without alignment.

| Column | Shows | Tooltip |
|---|---|---|
| Stage | the scan, as a SPEC command line | |
| Pick | the statistic the motor was moved to: CEN, pl_COM or pl_xMAX | the target position |
| Before → after | the motor before the scan, and after the move and any `set` | |
| CEN | the centre at half maximum | COM and the position of the maximum |
| FWHM | the full width at half maximum | |

| Column | Shows | Tooltip |
|---|---|---|
| Set | the value given to `set`, empty when the stage did not set | how far the user offset moved |

The statistics are splot's (`CEN`, `pl_COM`, `pl_xMAX`), the ones SPEC's macros read; bl5a-qs and the GUI compute them with the same code.

## Alignment history



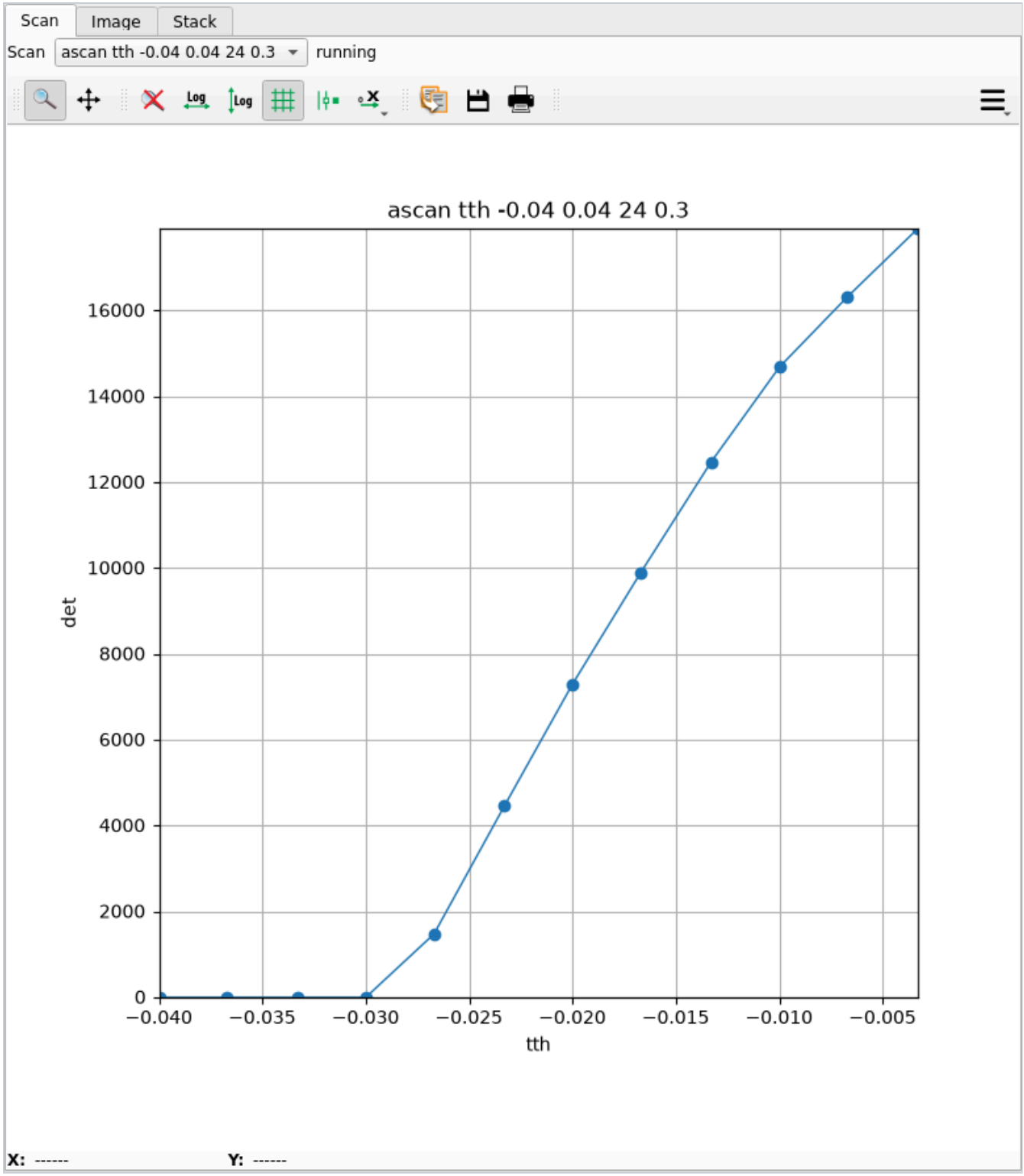

*The alignment history*

One row per finished run that carried alignment stages (or, for the coin cell, its recorded positions):

| Column | Shows |
|---|---|
| Time | when the run started |
| Series | the series it was queued in |
| Plan | `align_coin`, `align_film` or `insitu_step` |
| T | the `insitu_step` setpoint, °C |
| sy, shold | where the run left them |
| Δth0 | how far this run moved th's zero |
| Exit | `success`, `fail` or `abort`; its tooltip gives the reason a run did not succeed |

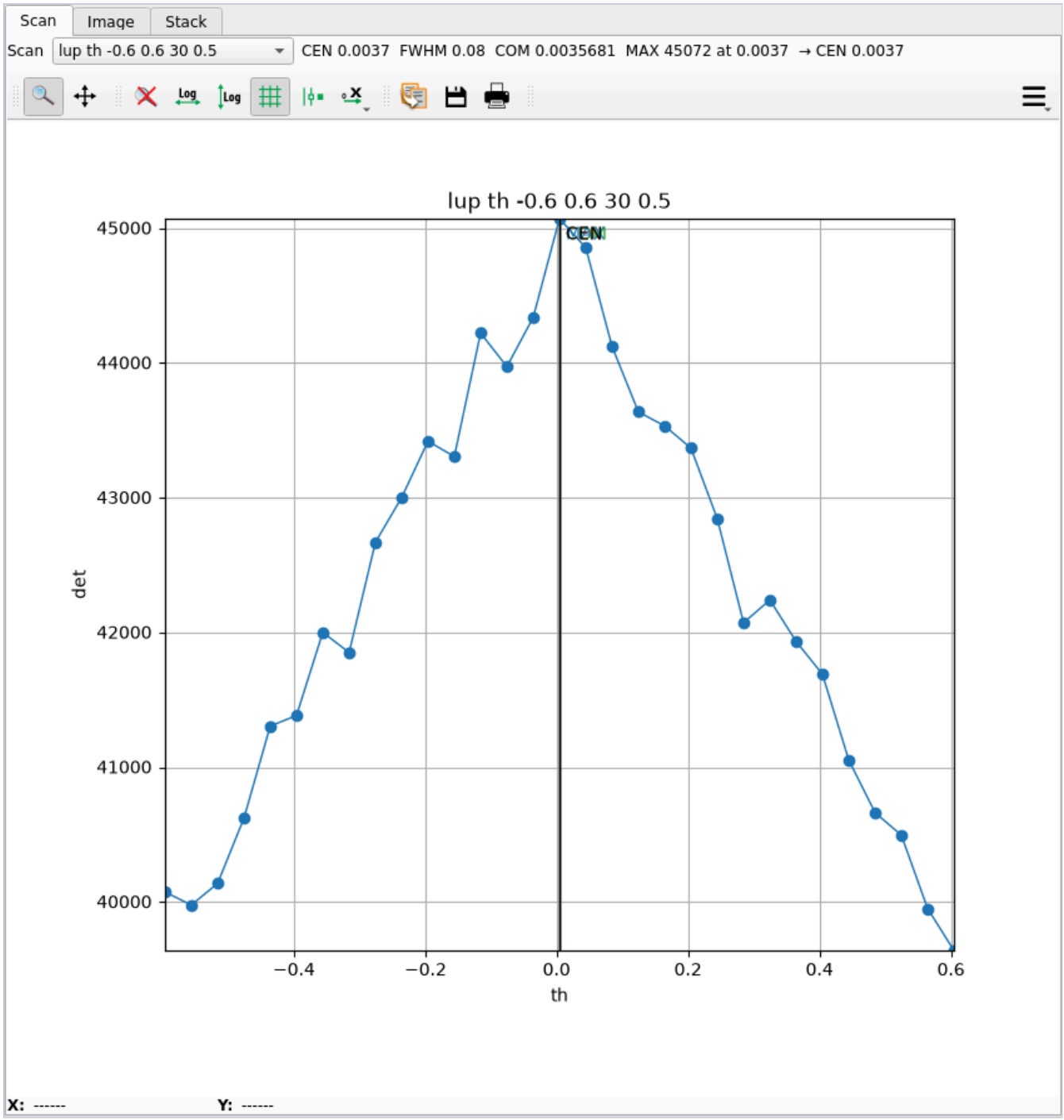
Δth0 is the sum of this run's `set th` corrections, not th0 itself: SPEC's dial position is not published, so the absolute offset is not known to the GUI.

## Scan tab




*A scan being drawn*

A scan is drawn point by point as it runs, at most five times a second. A fly scan runs in SPEC, which keeps its sampling timing; each of its points is drawn as SPEC takes it (spec-ioc passes on SPEC's `SCAN_PT` events), and is the counts SPEC took in one sample, at the middle of the motor's travel over it. When SPEC finishes, its full table of the scan adds any point not seen yet; a table that does not begin with the points already drawn fails the scan.



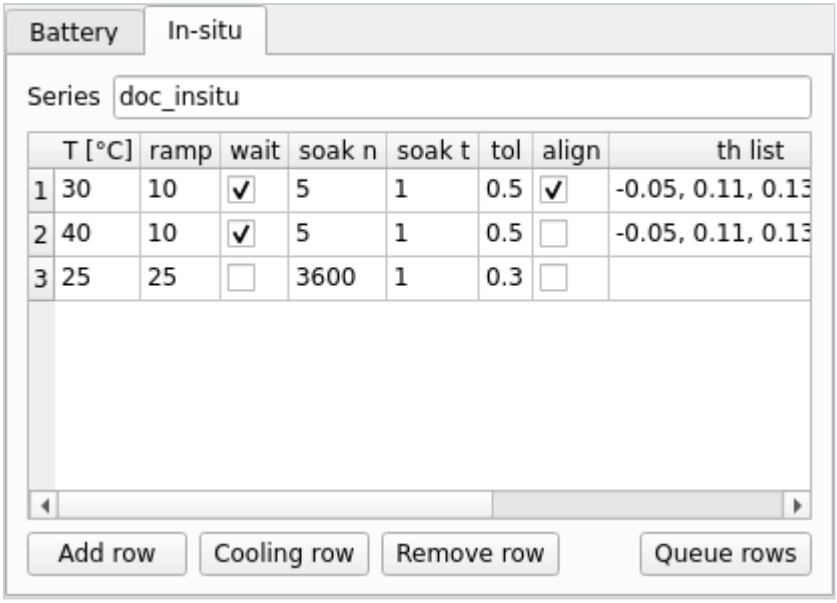

*A finished scan with its statistics*

When the scan ends, the line above the plot gives CEN, FWHM, COM and the maximum and its position, and markers are drawn: CEN red, COM green, MAX blue, and the stage's target black with the pick's name. In the picture the stage picked CEN, so the black target lies on the red CEN line. A scan that did not end in success shows its exit status and why; the whole line is in its tooltip when it does not fit.

The **Scan** list keeps this window's last 100 scans; pick one to look at it again. Every scan, from any window or none, is also appended to the SPEC-format file of its item's data folder, `<series>.spec` (see Data and install.md section 4.5), for splot or PyMCA. The toolbar is silx's: zoom, pan, log axes, grid, save, print.

## In-situ (film against temperature)

The In-situ form is SPEC's `myinsitu` as a table. Each row is one `insitu_step`; the rows run in order and their frames go to one series.




*The in-situ table: 30 °C, 40 °C, then the cooling row*

| Column | Default | What it is |
|---|---|---|
| T [°C] | empty | the setpoint (`settemp T`); empty leaves the temperature alone |
| ramp | empty | °C/min (`set_ramp`) |
| wait | off | wait until T is reached (`changec T` instead of `settemp T`) |
| soak n | empty | `loopscan` points while soaking |
| soak t | empty | seconds per soak point |
| tol | 0.3 | with wait: how close to T counts as reached, °C (`changec`'s `DELTA_K`) |
| align | on | run `align_in.mac` (the Thin film alignment) before the frames |
| th list | -0.05, 0.11, 0.13, 0.15 | one frame at each th, comma separated; empty takes no frames |
| texpo | -70000 | seconds (> 0) or monitor counts (< 0) |
| name | empty | the frames' file name |

**Series** names the data folder and stack of every row; empty means the first row's name.

One row runs, in order:

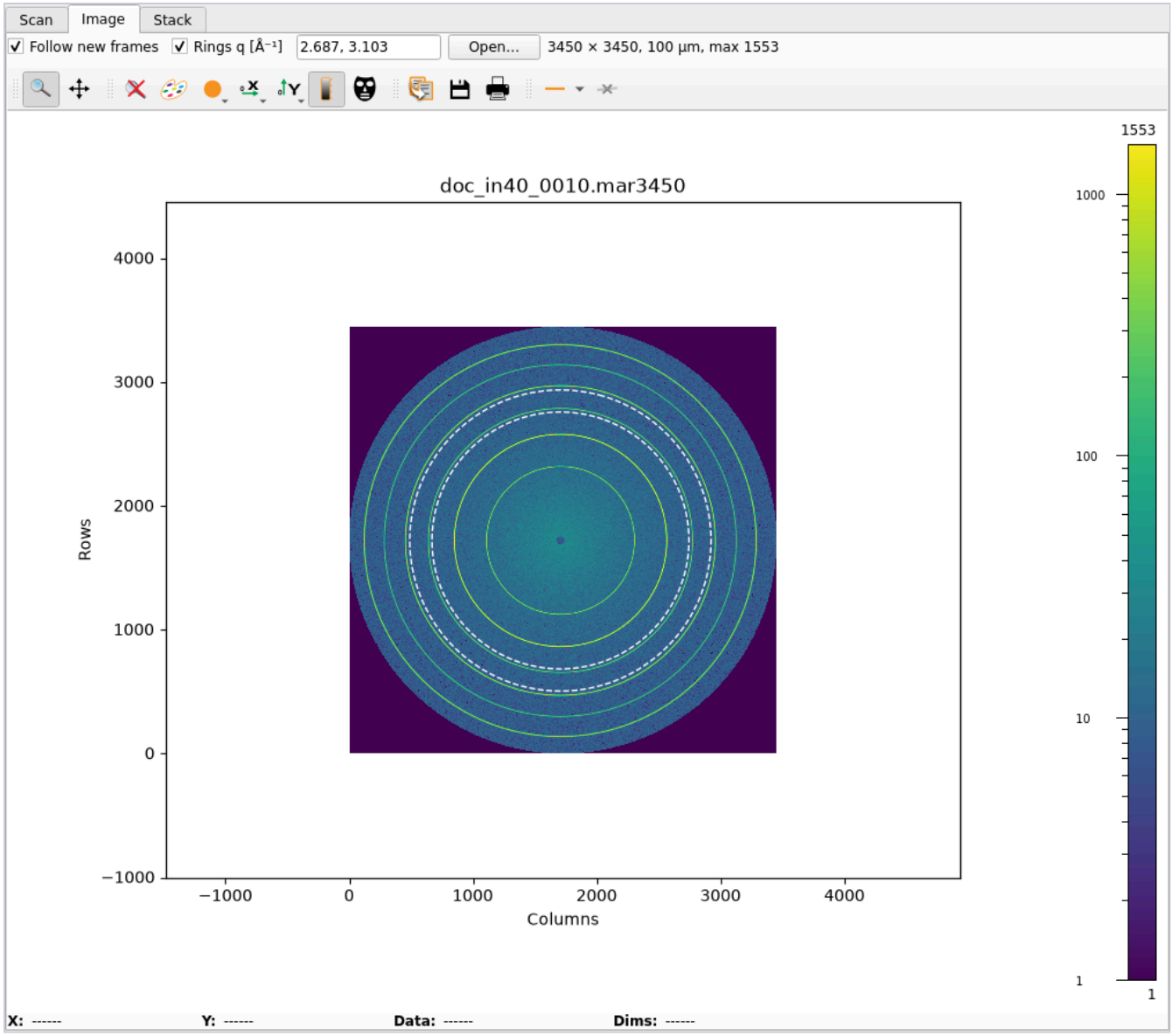
1. Temperature, only the parts filled in: `qdo` of the beamline's `temperature.mac`, `set_ramp <ramp>`, then `settemp T`, or with wait `changec T` with `DELTA_K` set to tol, then `loopscan n t`.
2. With align: the Thin film alignment above.
3. With a th list: `mymar_in`. If tth is below 40 it first closes the shutter and moves through `uan 45 0` to the in-situ positions. Then `uan 81 0`, `counters 2 3`, `fi 1 0 0 0`, `ic_range 5`, `ic_offset 0`, `mar_ip7 25 440 42.4 340 81.8 1.2 -6.8`, 5 s, `scell` to 0.9, a frame at each th, `ic_offset 1`, `fi 1 0 1 1`, back to tth 42 with the mar345 out of the way, and `uan 1 0`.

| Button | What it does |
|---|---|
| Add row | an empty row with the defaults |
| Cooling row | `myinsitu`'s last step: ramp 25 °C/min to 25 °C, no wait, soak 3600 points of 1 s, no alignment, no frames |
| Remove row | removes the selected rows |
| Queue rows | checks every row, then queues them all in order |

The checks: name 1 to 27 characters without spaces or quotes (only when the th list is not empty); texpo not 0; ramp above 0; wait needs a T; with wait, tol above 0 (without wait tol is not used); soak needs both n (a whole number, at least 1) and t (above 0); a row with no T, ramp or soak, no alignment and no th list does nothing and is refused. A bad row opens an "In-situ" box that names the row and the field; nothing is queued then.

The cooling row soaks for an hour. Queue it last.

## Image tab

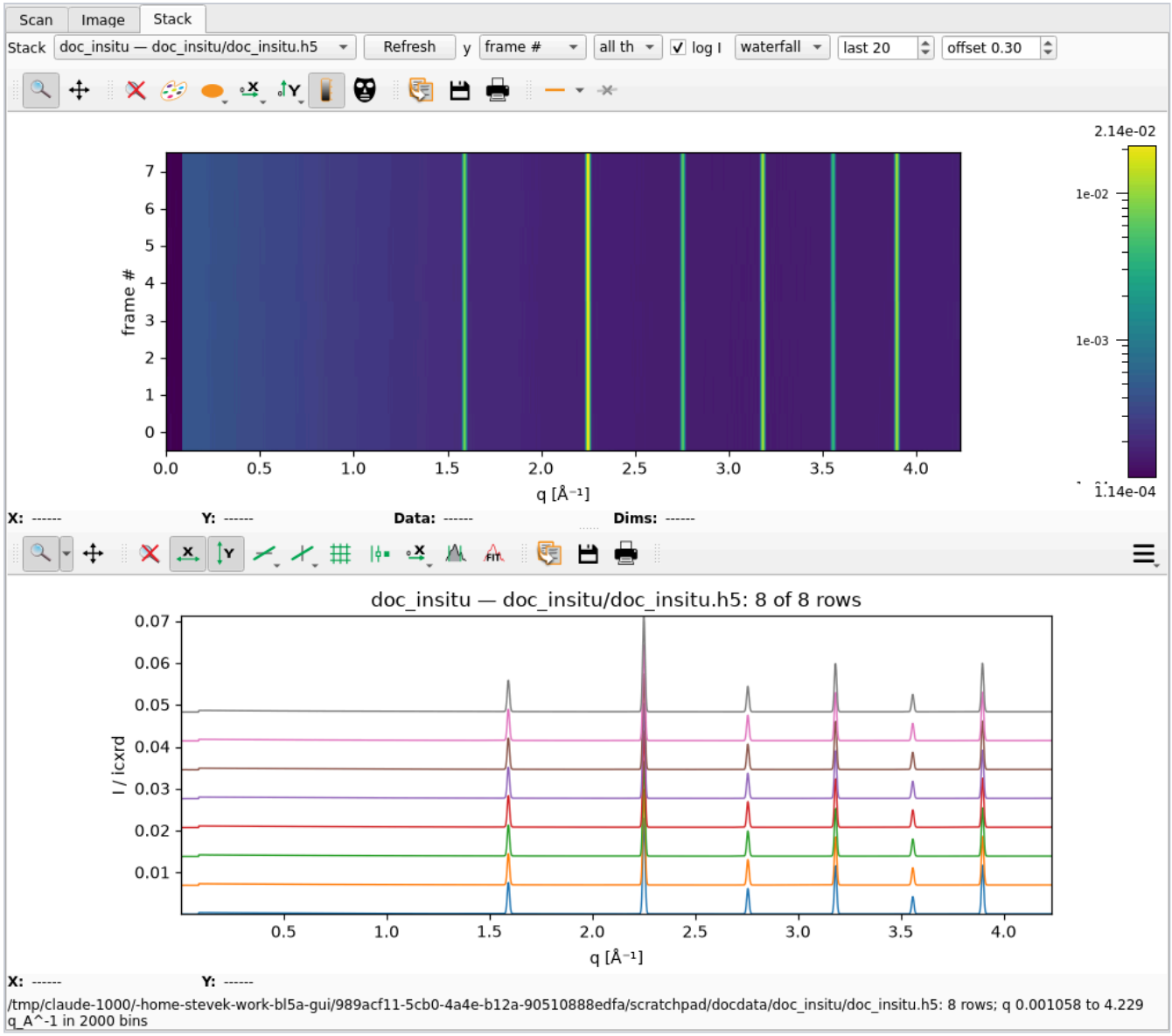



*The last frame with the reference rings*

- **Follow new frames**: each new frame is loaded as it is written. The file is read off the GUI's thread, from the path the mar345 IOC reports, so that path must be readable on the GUI's computer.
- The image is drawn on a log colour scale; zero counts are drawn as 1.
- **Rings q [Å⁻¹]**: comma-separated q values, drawn as dashed white circles where the integrator's geometry puts them. The defaults are Al 111 and Al 200 (2.687 and 3.103). The simulated film in the picture has no aluminium, so its rings fall elsewhere.
- **Open...** loads any mar345 file.
- The line after the buttons gives the image size, the pixel size and the largest count, or the reason a file could not be read.

The rings use the integrator's PONI, which the GUI asks the integrator for. The integrator uses one PONI for every frame, so the rings and the stacks are right only for the detector position that PONI was measured at (see install.md).

## Stack tab




*A series stacked: the heatmap and the waterfall*

The integrator integrates every frame and appends it to its series' stack in the item's data folder. A stack is one series on one q axis (see Integration). The tab shows one stack:

- the heatmap: q across, one row per frame (or by time), log colour;
- the profiles: I / icxrd against q for the last N rows.

| Control | What it does |
|---|---|
| Stack | every stack the integrator holds, as `series — folder/file`, in path order; at start the last one is shown |
| Refresh | asks the integrator again for its settings and its stacks |
| y | `frame #`, or `time [min]` from the first frame |
| all th / th v | all frames, or only those taken at one th |
| log I | log colour scale (default on) |
| overlay / waterfall | profiles drawn on top of each other, or stepped up |

| Control | What it does |
|---|---|
| last N | how many of the newest rows the profiles show (default 20) |
| offset | the waterfall step, as a fraction of the rows' median maximum (default 0.3) |

New rows arrive by themselves. A new stack shows up in the list when its first frame is integrated; the tab stays on the stack shown, or shows the new one when none was shown.

The line at the bottom gives the stack's file, its number of rows and its q axis. It reads "integrator: not answering" when the integrator does not reply. The integrator's settings are in the Integration box.

The integrator reads a folder's stacks once, at its start or at the folder's first row. A stack file deleted on disk while it runs stays in the list until the series' next frame, which starts the file again.

## SPEC console, Commands, Messages




*The SPEC console*

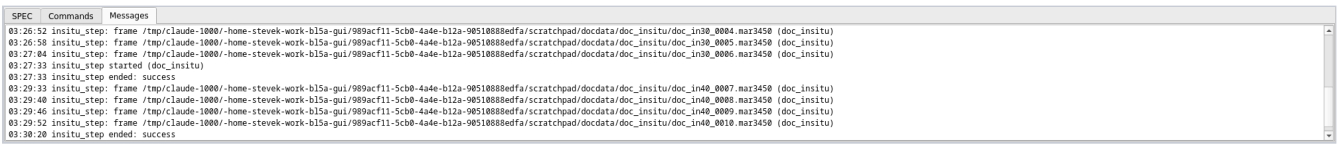
**SPEC** shows what SPEC prints, as spec-ioc's `Log` PV carries it, the plans' output included. `--- N lines not shown ---` marks a burst longer than the PV holds between two updates; `--- spec-ioc restarted ---` marks an IOC restart.

The entry sends one SPEC command (Enter or **Send**) when the queue is idle (see Who drives SPEC). The command is echoed as `HH:MM:SS SPEC> command`, followed by SPEC's reply. A command that does not end OK adds a line with its status. **^C** interrupts the command sent from here; it never touches a plan's command.




*The commands the plans sent*

**Commands** lists every SPEC command a plan sent, with the time and the plan, and its reply indented below.



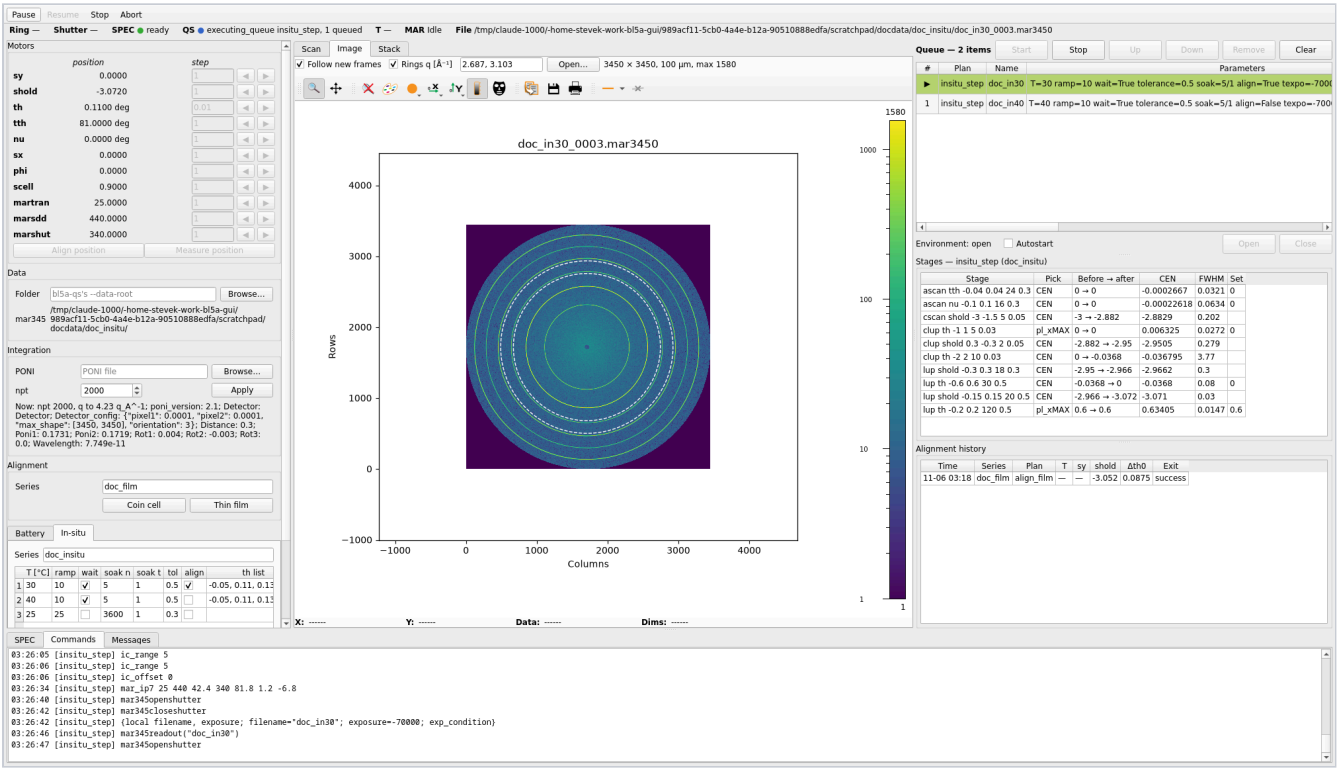

*The GUI's messages*

**Messages** is the GUI's own log: each queue request and its result, each plan's start, each queue item's end with its exit status (and, when it did not succeed, why), each frame with its series, and errors from bl5a-qs and the integrator. A request bl5a-qs refuses also opens a "bl5a-qs" box.

## Walkthrough: a battery cycle

1. Mount the coin cell. Check the **Data** folder. In **Alignment**, type a series (for example the cell's name) and click **Coin cell**.
2. Click **Start** in the queue. The Scan tab draws the sy and shold fly scans; the history gets a row with the sy and shold the run left.
3. With the queue idle, type `umv tth 81` in the SPEC console.
4. On the **Battery** form: a file name, Texpo, **Repeat** with N and Ti (or **Single shot**), the geometry and the same series, so the frames go to the alignment's folder. **Add to queue**, answer **Yes** to the beamstop question, then **Start**.
5. Start the cycler from its own computer.
6. Each frame shows on the **Image** tab and is added to the **Stack** tab; choose `time [min]` for y to follow the cycle.
7. To end before N frames, click **Stop after frame**.

## Walkthrough: a film against temperature




*An in-situ run: the frame, the queue, the stages and the commands*

1. Mount the film. Check the **Data** folder and, for the detector position `mymar_in` uses, the **Integration** PONI. On the **In-situ** form, type the series.
2. Fill one row per temperature: T, ramp, wait and tol, a soak if wanted, align, the th list, texpo and the file name. Add the **Cooling row** last.
3. **Queue rows**, then **Start**.
4. Each row sets the temperature, aligns (Stages and Scan tabs) and takes its frames (Image and Stack tabs). The history gets a row per aligned step with its T.
5. On the **Stack** tab, pick one th (for example `th 0.13`) to compare the same angle across temperatures.

## AI control (MCP)

`gui/run_mcp.py` gives Claude Code or Codex the GUI's controls as MCP tools (setup: install.md). The client starts it through `deploy/bl5a-mcp` when it starts in the `bl5a_gui` directory. It builds queue items with the forms' own code, passes the same gate as the buttons and writes only the PVs the GUI writes, so an item the AI queues is the item the form would queue. It runs beside the GUI, and the GUI shows what the AI queued and runs.

Ask in plain words, for example "queue a thin-film alignment for series film3 and start it", "show me the last frame" or "what did the last fly scan find?".

| Tools | What they do | In the GUI |
|---|---|---|
| `status` | the queue's and SPEC's state, and `allowed`: the actions open now | status line; which buttons are live |
| `queue`, `history` | the waiting items; the finished ones with their exit status | Queue |
| `motors` | position, step and unit of each motor | Motors |
| `scans`, `scan`, `alignment` | the scans with their peak values; the stages and the alignment history | Scan tab, Stages, Alignment history |
| `messages`, `spec_log`, `commands` | the three logs | Messages, SPEC, Commands |
| `frame_image` | the last frame, or a named one, as a picture | Image tab |
| `integration`, `stacks`, `stack` | the integrator's PONI and npt; the stacks; one stack's rows | Integration, Stack tab |
| `queue_alignment`, `queue_mar_series`, `queue_insitu` | queue items | Alignment buttons, Battery form, In-situ form |
| `queue_remove`, `queue_move`, `queue_clear`, `queue_start`, `queue_stop`, `queue_autostart`, `environment_open`, `environment_close` | edit and run the queue | the queue's buttons |
| `pause`, `resume`, `stop`, `abort`, `stop_after_frame` | end or hold the running plan | Run buttons, Stop after frame |
| `tweak` | move a motor by a step | Motors ◀ ▶ |
| `spec_command`, `spec_result`, `spec_abort` | one SPEC command, its result, ^C | SPEC console |
| `integration_set` | a PONI file and npt for the next frames | Integration Apply |

The rules are the GUI's:

- An action that is not open now is refused with the reason, where the button would be grey: `pause: not open now (qs idle; …)`. The tools ask bl5a-qs first, so each one sees what the previous one did.

- `data_folder` left out uses the GUI's Data folder; `""` uses bl5a-qs's `--data-root`; an absolute path is used as it is. The data then goes to `<folder>/<series>`.
- `queue_mar_series` needs `beamstop_in_place: true`, the GUI's beamstop question. The server tells the AI to ask you first, and to ask before `abort` and `queue_clear`. Your MCP client's permission prompt is the other guard: allow only the read tools without a prompt.
- `pause` then `resume` fails the plan, as in the GUI. `stop_after_frame` ends a series after the frame being taken; asked before the first frame starts, the series stops with none.
- `spec_command` waits up to `wait_s` (default 30 s); a longer command comes back as running, and `spec_result` gives its end. What SPEC prints (the table of `wm`, for example) goes to the SPEC log, not the reply: read it with `spec_log` from the `spec_log_since` the result gives.
- The logs hold what happened since the server started. Each read gives `last`; pass it as `since` next time to get only the new lines.

As on the Battery form, move tth to 80° or more before a standard `queue_mar_series` (Battery measurements); a thin-film alignment leaves it near 1.2. Use `tweak`, or `spec_command` with `umv tth 81` at the beamline.

## When something is wrong

| What you see | What it means | What to do |
|---|---|---|
| QS ● not answering | bl5a-qs does not reply on `--qs-control` | `systemctl --user status bl5a-qs`, its journal (`journalctl --user -u bl5a-qs`), or the address |
| SPEC — | spec-ioc's PVs do not connect | `systemctl --user status bl5a-spec-ioc`; check Channel Access to its host |
| SPEC ● disconnected | spec-ioc runs but SPEC's server does not answer it | start SPEC with `-S`; check the host and port in `SPEC=` |
| Motor buttons and the console stay grey | a plan runs, SPEC is busy, or a console command runs | wait, or stop the queue |
| "Battery", "In-situ" or "Alignment" box | a field breaks a rule, a series that is not a folder name included | the box names the field and the rule |
| "bl5a-qs" box | bl5a-qs refused a request | the box gives its reason |
| Messages: "... ended: fail: ..." and Autostart unticked | the item failed; the text after `fail:` says why | the last SPEC command and its reply are at the bottom of Commands |
| Image info line shows a file error | the frame's path is not readable here | mount the mar345 data directory at the same path on this computer |
| Stack: "integrator: not answering" | bl5a-integrator is not running or not at `--integrator-rep` | `systemctl --user status bl5a-integrator`; Refresh |
| "Data" box | the Data folder is not an absolute path, holds a space, or with the series is over 200 bytes | the box gives the rule |
| Data: mar345 folder "missing" | the mar345 IOC does not see the item's folder | mount the data root at the same path on the IOC's host |
| Messages: "... ended: fail: device state error: data folder: ..." | the data root does not exist on bl5a-qs's host | create it, or fix the Data folder and queue again |

| What you see | What it means | What to do |
|---|---|---|
| Integration: "Refused: …" | the integrator could not use the PONI or npt | the text after "Refused:" says why |
| T reads "—" | no temperature PVs are configured | the temperature is set by SPEC's macros; read it in SPEC |